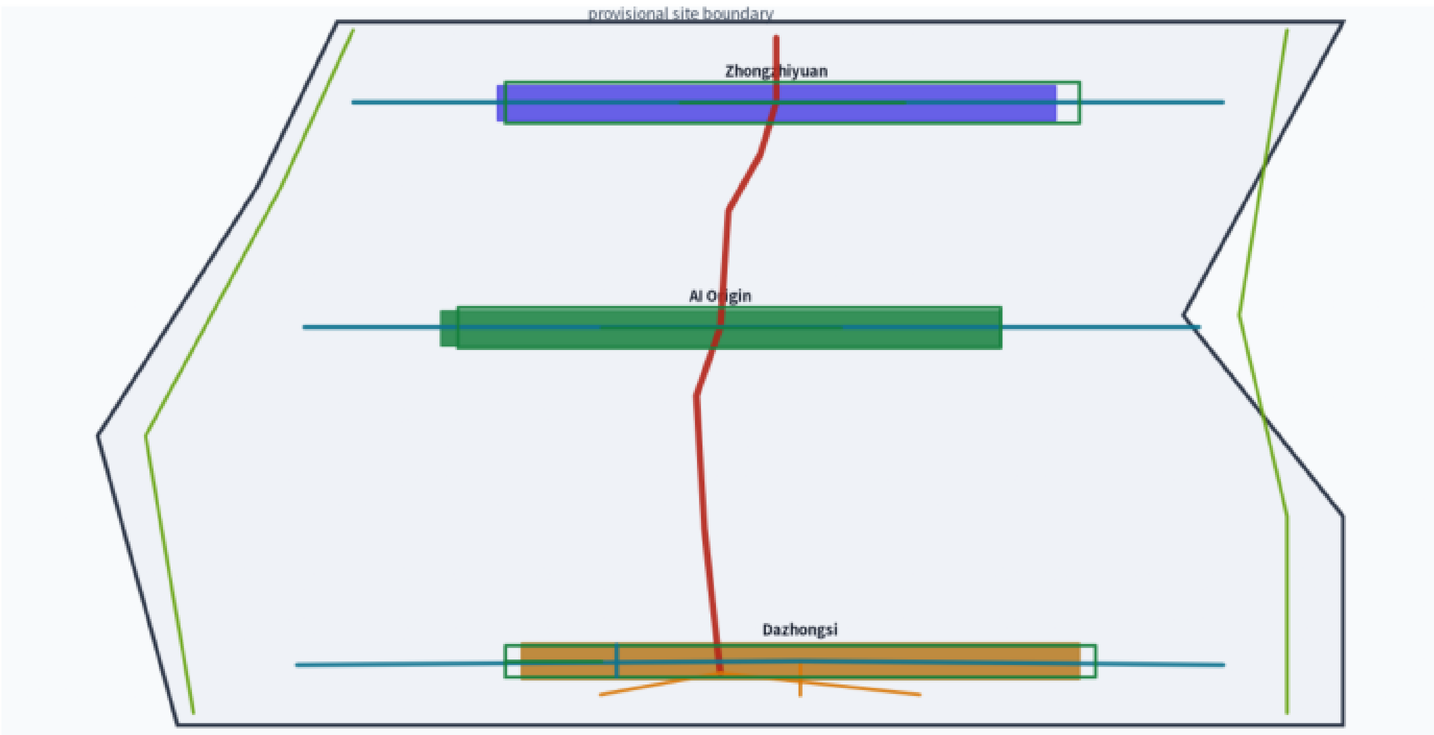
A3 Booklet

Dazhongsi AI Cluster · Bio-Physarum Network Renewal

智脉共生：大钟寺AI产业集聚区生物黏菌路网更新概念方案

Evidence chain and package



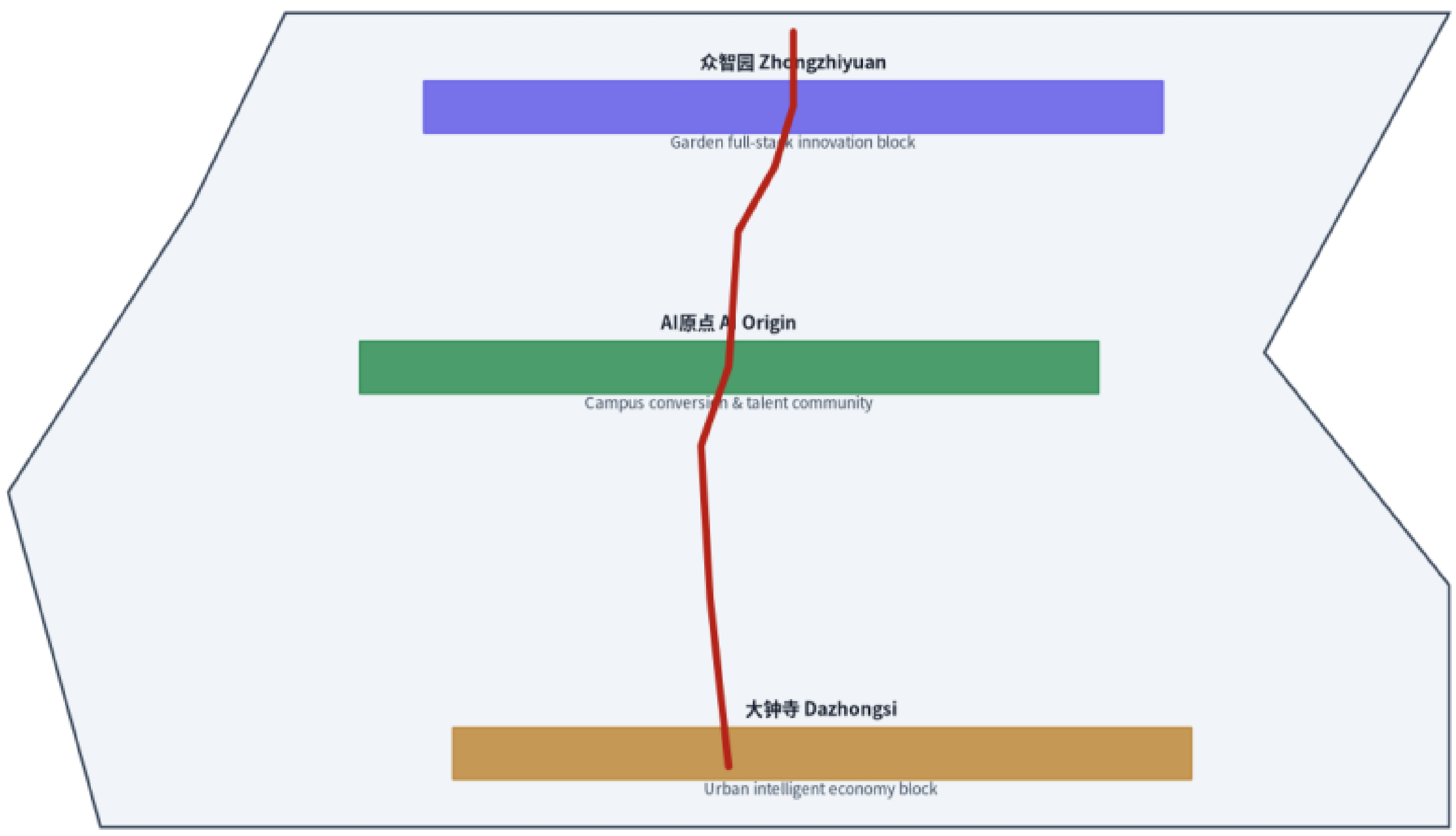
Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective | Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective f3≡f2 (degenerates to cost; no independent site heritage conclusion)

Three-level scope and spatial structure

Coordinated study scope	$43.6\ km^2$	AI ecosystem / three zones two wings / future urban form
Overall design scope	$11.4\ km^2$	Urban renewal framework / land use / transport & utilities
Key area scope	368.4 ha	Three key detailed-design areas

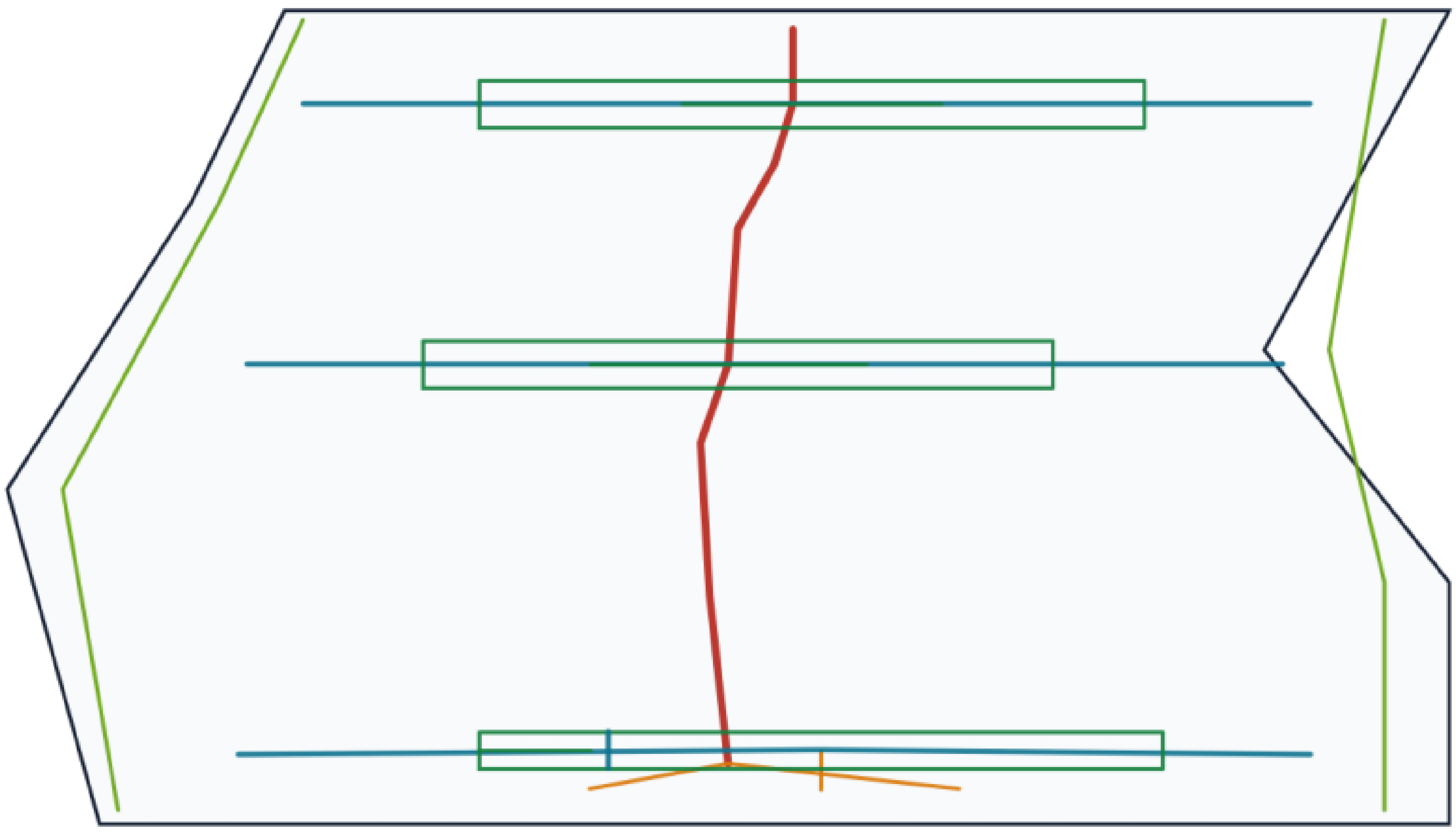
Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective

Key areas and Physarum spine

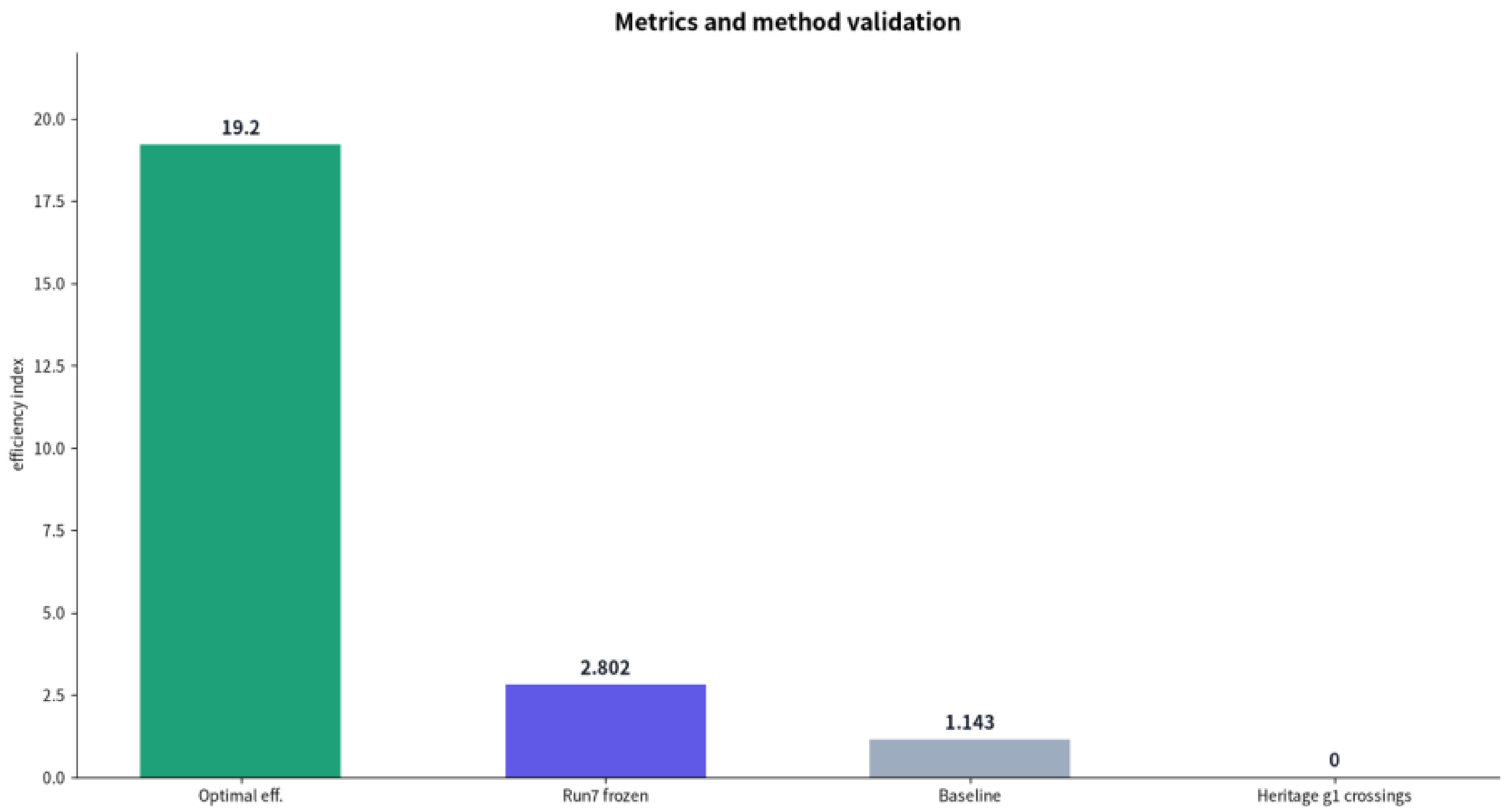


Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective

Mobility and blue-green public space



Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective



Plan03 urban-integration UDS 80.34 recommended; heritage objective f3≡f2 degenerates to cost — '0 heritage crossings' is g1 hard-constraint behavior only, not an independent site heritage conclusion
Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective f3≡f2 (degenerates to cost; no independent site heritage conclusion)